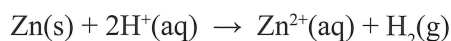
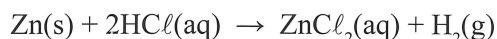


Set 30: Stoichiometry and gas volumes

Example

Determine the volume of hydrogen (measured at S.T.P.) is produced when 45.6 g of zinc metal is added to excess hydrochloric acid.

- (i) Write an equation for the reaction



- (ii) Calculate the number of moles of zinc

$$n(\text{Zn}) = \frac{m}{M}$$

$$= \frac{45.6}{65.38}$$

$$= 0.697 \text{ mol}$$

- (iii) From the equation determine the number of moles of hydrogen

$$n(\text{H}_2) = n(\text{Zn})$$

$$n(\text{H}_2) = 0.697 \text{ mol}$$

- (iv) Calculate the volume of hydrogen

$$V = n \times 22.71$$

$$= 0.697 \times 22.71$$

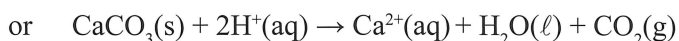
$$= 15.8 \text{ L}$$

Notes

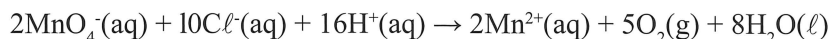
Set 30: Exercises

1. Calculate the volumes of gas at S.T.P. produced in each of the following reactions:

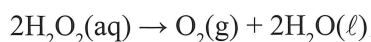
- (a) When 0.250 moles of CaCO_3 is treated with excess hydrochloric acid:



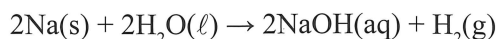
- (b) When 0.150 moles of potassium permanganate reacts with excess concentrated hydrochloric acid according to the equation



- (c) When 0.0300 moles of hydrogen peroxide disproportionates according to the equation



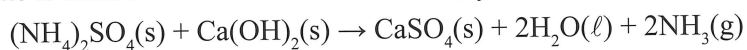
2. If 4.60 g of sodium reacts with water, calculate the volume of hydrogen produced at S.T.P.:



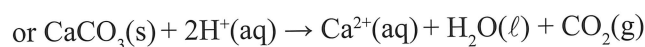
Set 30: Stoichiometry and gas volumes

Notes

3. Calculate the volume of ammonia produced at S.T.P. when 22.8 g of ammonium sulfate is heated with excess moist calcium hydroxide:

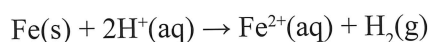


4. When 25.9 g of impure limestone was treated with excess hydrochloric acid, 5.6 L of carbon dioxide was produced at S.T.P.:



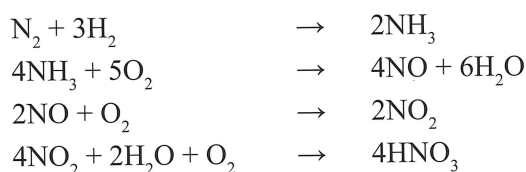
Calculate the percentage purity of the limestone.

5. A 15.3 g piece of steel, containing only iron and carbon, was treated with plenty of excess hot acid to form 6.03 L of hydrogen at S.T.P.:



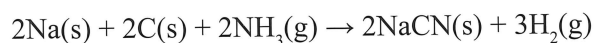
Calculate the percentage of iron in the steel.

6. The reactions involved in the manufacture of nitric acid can be represented as follows:



Calculate:

- (a) the mass of nitrogen, and
 - (b) the volume of nitrogen at S.T.P. required to manufacture 25.0 kg of nitric acid.
7. Methane (CH_4) burns in oxygen to form carbon dioxide and water. Write a balanced equation for the process.
If 2.50 L of gaseous water is produced at S.T.P., calculate the volumes of:
- (a) carbon dioxide produced, and
 - (b) methane consumed at the same temperature and pressure.
8. Sodium cyanide, which is used in the extraction of gold, can be made from sodium metal, carbon and ammonia:



Calculate the mass of sodium cyanide that would be formed by reacting 0.500 tonne of sodium metal with 762 kL of ammonia gas at S.T.P. in the presence of excess carbon.